Beechcraft Baron 58 (BE58)

2× Continental IO-520-C, 285 hp (IO-550-C, 300 hp from 1984) · 6 seats · Typical Baron 58 (5,400–5,500 lb) POH procedures; IO-520 and IO-550 variants share speeds, power settings differ slightly.

V-speeds & limits

Vr	**84 KIAS (liftoff)**
Vmca	**84 KIAS**
Vsse	**88 KIAS (intentional one-engine-inop)**
Vx	**92 KIAS**
Vxse	**100 KIAS**
Vyse	**101 KIAS (blue line)**
Vy	**105 KIAS**
Vfe	**152 KIAS (approach 15°) / 122 KIAS (full 30°)**
Vlo / Vle	**152 KIAS**
Va	**156 KIAS (5,500 lb)**
Vno	**195 KIAS**
Vne	**223 KIAS**
Vs / Vso	**84 / 75 KIAS (max weight)**
Vref	**95 KIAS final (full flaps); 85–90 KIAS over threshold**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Passenger briefing	**COMPLETE (belts, cabin / aft doors, exits, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door / aft cabin doors / baggage doors	**CLOSED & LATCHED**
Parking brake	**SET**
Avionics master	**OFF**
Landing gear switch	**DOWN**
Throttles / propellers / mixtures	**IDENTIFY – throttles closed, props high RPM, mixtures idle cutoff**
Fuel selectors	**ON both (LEFT – ON, RIGHT – ON; not CROSSFEED)**
Fuel quantity	**CHECK both sides**
Cowl flaps	**OPEN**
Alternate static source	**OFF (normal)**
Battery & alternator switches	**ON**
Annunciators	**PRESS TO TEST**
Landing gear indicator	**GREEN (down)**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR (both sides)**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Throttle	**FULL OPEN**
Auxiliary fuel pump	**HI until fuel flow peaks (3–5 sec), then OFF**
Throttle	**1/4 OPEN**

▶ **Engine hot or flooded?**

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Magneto / start switch	**START (release when engine fires)**
Throttle	**900–1000 RPM**
Oil pressure	**10 psi within 30 sec, rising to 30–60 psi**
Second engine	**REPEAT start sequence**
Alternators	**CHECK both charging (loadmeters / voltmeter)**
External power	**DISCONNECTED (if used)**
Avionics master	**ON**
Engine instruments	**CHECK both engines**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeters, HSI / DG, turn coordinator during turns)**
Avionics / GPS	**SET**
Autopilot / electric trim	**CHECK & OFF**
Trims (elevator, rudder, aileron)	**SET for takeoff**
Fuel selectors	**CHECK ON both**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selectors	**ON both (feel for detents)**
Mixtures	**FULL RICH (lean per placard above approx 5,000 ft elevation)**
Throttles	**1700 RPM (one engine at a time)**
Magnetos	**CHECK L & R each engine (150 RPM max drop, 50 RPM max difference)**
Propellers	**CYCLE 2–3 times (RPM drops, then returns)**
Propellers – feather check	**1500 RPM, pull to FEATHER momentarily – RPM drops (approx 300 RPM); return before 1000 RPM**
Engine instruments	**CHECK both (oil pressure 30–60 psi, oil temp green, fuel flow steady)**
Alternators	**CHECK both charging**
Gyro suction / pressure	**CHECK in green**
Throttles	**IDLE check, then 1000 RPM**
Auxiliary fuel pumps	**OFF**
Cowl flaps	**OPEN**
Propellers	**HIGH RPM**
Flaps	**UP (check indicator)**
Trims	**SET takeoff range**
Landing gear switch	**DOWN – IDENTIFY (gear and flap switches are on opposite sides of the quadrant)**
Pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Doors & windows	**LATCHED**
Takeoff briefing	**COMPLETE (Vmca 84 / Vr 84 / Vyse 101; abort below liftoff; engine-out plan)**

After Takeoff / Climb

Landing gear	**UP (positive rate, no usable runway, below 152 KIAS)**
Gear indicator lights	**OUT (gear up and locked)**
Flaps	**UP**
Climb power	**25 in Hg / 2500 RPM (cruise climb) – full throttle / 2700 RPM permitted**
Mixtures	**LEAN per POH climb fuel-flow schedule above approx 5,000 ft**
Cowl flaps	**OPEN (CHT / oil temp in green)**
Auxiliary fuel pumps	**OFF (LO if fuel flow fluctuates)**
Airspeed	**120 KIAS (cruise climb)**
Engine instruments	**CHECK both**

Cruise

Cruise power	**SET (2300–2500 RPM, 23–25 in Hg typical)**
Mixtures	**LEAN per POH cruise table / EGT**
Cowl flaps	**CLOSED (as required)**
Propellers	**SYNCHRONISE**
Fuel quantity / flow	**CHECK both sides against plan**
Fuel selectors	**ON both (CROSSFEED only for one-engine-inop balance in level flight)**
Engine instruments	**CHECK both (oil temp 240°F max)**
Altimeter / heading	**CHECK**

Descent

Altimeters	**SET**
Fuel selectors	**ON both**
Mixtures	**ENRICH progressively**
Power	**REDUCE gradually (avoid rapid cooling)**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 152 KIAS before gear / approach flaps**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selectors	**ON both**
Auxiliary fuel pumps	**OFF (LO if fuel flow fluctuates)**
Mixtures	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 152 KIAS**
Landing gear	**DOWN – GREEN light (identify gear switch, not flap switch)**
Flaps	**APPROACH below 152 KIAS; DOWN on final below 122 KIAS**
Propellers	**HIGH RPM (on final)**
Airspeed	**95 KIAS final; 85–90 KIAS over threshold**
Landing lights	**ON**
Landing gear	**GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (look and identify the flap switch before moving)**
Cowl flaps	**OPEN**
Auxiliary fuel pumps	**OFF**
Pitot heat	**OFF**
Landing lights / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttles	**1000 RPM (cool-down)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Mixtures	**IDLE CUTOFF**
Magnetos	**OFF (after propellers stop)**
Battery & alternator switches	**OFF**
Control lock	**INSTALL**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT OUT light, one loadmeter zero or voltage below 24 V

Affected alternator switch	**OFF, then ON (reset)**
Alternator field circuit breaker	**CHECK / RESET once**

▶ Alternator restored?

Yes:

Loadmeters / voltmeter	**MONITOR both**

No – one alternator failed:

Affected alternator switch	**OFF**
Electrical load	**REDUCE to within remaining alternator capacity**
Deferred: One alternator inoperative – electrical load reduced, land as soon as practical.	**NOTED**

No – both failed:

Both alternator switches	**OFF**
Non-essential electrical load	**REDUCE (battery only, limited endurance)**
Landing gear	**PLAN early extension while battery power remains (hand crank if needed)**
Deferred: Both alternators failed – battery only; extend gear early, land as soon as possible.	**NOTED**

Low Oil Pressure (one engine)

Oil pressure below 30 psi in flight

Oil pressure / oil temperature	**CHECK and cross-check the affected engine**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

Affected engine power	**REDUCE to minimum required**
Engine	**SHUT DOWN and FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**
Deferred: Low oil pressure – precautionary engine shutdown; single-engine landing, land as soon as possible.	**NOTED**

No – gauge suspect:

Flight	**CONTINUE, monitor oil temperature and pressure**
Deferred: Oil pressure indication suspect – maintenance check required.	**NOTED**

High Oil / Cylinder Head Temperature (one engine)

Affected engine cowl flap	**OPEN**
Affected engine mixture	**ENRICH**
Airspeed	**INCREASE (lower pitch attitude)**
Affected engine power	**REDUCE**

▶ Temperature returning to green?

Yes:

Flight	**CONTINUE, monitor**

No:

Oil pressure	**MONITOR for drop**
Engine	**CONSIDER precautionary shutdown and feather if altitude and terrain allow**
Deferred: High engine temperature – land as soon as practical, cooling / baffling check required.	**NOTED**

Rough Running / Fuel Flow Fluctuation (one engine)

Affected engine auxiliary fuel pump ..... **LO (HI only if fuel flow very low – may flood engine at low power)**

Affected engine mixture ..... **ADJUST for smooth running**

Fuel selector ..... **CHECK ON; CROSSFEED from other side if fuel quantity suspect**

Affected engine magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (CHT, EGT, fuel flow)**

▶ Engine running smoothly?

Yes:

Deferred: Rough running cleared – leave aux pump LO if used; report to maintenance. ..... **NOTED**

No:

Affected engine power ..... **REDUCE to smoothest setting – feather if power unusable (Engine Failure in Flight)**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Flap Failure / Asymmetry

Flap switch ..... **NEUTRAL / stop – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – add 10 KIAS to approach speed, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Landing Gear Fails to Retract / Extend Normally

Airspeed ..... **BELOW 152 KIAS**

Landing gear circuit breaker ..... **CHECK / RESET once**

Landing gear switch ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear switch ..... **DOWN – leave down**

Airspeed ..... **BELOW 152 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range (negative single-engine climb), land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Manual Gear Extension (Hand Crank).

Cabin / Aft Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 120 KIAS**

Belts / loose items ..... **SECURE**

Do not attempt to close a door in flight. Expect increased noise and some buffet; the aft cabin doors may trail open.

Deferred: Door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

Gyro Suction / Pressure or Attitude Indicator Failure

Suction / pressure gauge ..... **CHECK (one pump failed – remaining pump maintains system)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**

Failed instruments ..... **COVER**

Autopilot ..... **OFF if coupled to failed gyro**

IMC ..... **EXIT / AVOID if both pumps or the attitude indicator failed**

Deferred: Gyro system failure – partial panel; prefer VMC approach. ..... **NOTED**

Propeller Overspeed

RPM above 2700 and not responding to prop lever

- Affected engine throttle — **REDUCE**
- Affected engine propeller — **LOW RPM (toward feather detent)**
- Airspeed — **REDUCE to bring RPM within limits**
- Affected engine oil pressure — **CHECK (governor is oil-operated)**

▶ RPM controllable at or below 2700?

Yes:

- Power — **SET minimum that keeps RPM in limits**
- Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. — **NOTED**

No:

- Affected engine — **SHUT DOWN and FEATHER (Engine Failure in Flight)**
- Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. — **NOTED**

Emergency procedures

Engine Failure During Takeoff

On the runway or just airborne with runway remaining

▶ Airborne?

No – below liftoff:

- Throttles (memory) — **CLOSED (both)**
- Brakes (memory) — **MAXIMUM – stop straight ahead**
- Mixtures — **IDLE CUTOFF**
- Fuel selectors — **OFF**
- Magnetos — **OFF**
- Battery & alternator switches — **OFF**

Yes – runway remaining:

- Throttles (memory) — **CLOSED (both)**
- Land (memory) — **STRAIGHT AHEAD, gear down**
- Brakes — **MAXIMUM**
- Mixtures / fuel selectors / magnetos — **OFF after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (84 KIAS) directional control cannot be maintained with full power on one engine – do not attempt to continue the takeoff below liftoff speed.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

- Mixtures (memory) — **FULL RICH**
- Propellers (memory) — **HIGH RPM**
- Throttles (memory) — **FULL**
- Flaps (memory) — **UP**
- Airspeed (memory) — **101 KIAS (Vyse) – never below 84 KIAS**

▶ Landing gear position?

Gear DOWN:

With gear down, climb performance on one engine is negative at most weights. Retract if a climb is required; otherwise land straight ahead.

- Landing gear (memory) — **UP (if committed to fly)**
- Inoperative engine (memory) — **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) — **VERIFY – retard, no change**
- Inoperative engine propeller (memory) — **FEATHER**
- Inoperative engine mixture (memory) — **IDLE CUTOFF**

Gear UP:

- Inoperative engine (memory) — **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) — **VERIFY – retard, no change**
- Inoperative engine propeller (memory) — **FEATHER**
- Inoperative engine mixture (memory) — **IDLE CUTOFF**

- Bank — **2–5° toward operating engine, ball half out**
- Trim — **RUDDER & AILERON for straight flight**
- Climb — **101 KIAS to safe altitude, then Engine Securing**
- Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. — **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

Mixtures (memory)	FULL RICH
Propellers (memory)	HIGH RPM
Throttles (memory)	FULL (as required)
Landing gear / flaps (memory)	UP
Airspeed (memory)	101 KIAS (Vyse) minimum
Inoperative engine (memory)	IDENTIFY – dead foot, dead engine
Inoperative engine throttle (memory)	VERIFY – retard, no change
Inoperative engine propeller (memory)	FEATHER
Inoperative engine mixture (memory)	IDLE CUTOFF
Bank / rudder	2–5° toward operating engine, ball half out
Trim	RUDDER & AILERON

▶ Cause obvious (fuel) and altitude available for a restart?

Yes – attempt air start:

Use the Air Start procedure. If unsuccessful, secure the engine.

No – secure engine:

Continue with Engine Securing, then divert to the nearest suitable airport.

Deferred: One engine inoperative – single-engine approach and landing, drift-down altitude check, no go-around once committed. ..... **NOTED**

Engine Securing (after feather)

Inoperative engine propeller feathered

Inoperative engine fuel selector	OFF
Inoperative engine auxiliary fuel pump	OFF
Inoperative engine magnetos	OFF
Inoperative engine alternator	OFF
Inoperative engine cowl flap	CLOSED
Propeller synchroniser	OFF (if fitted)
Electrical load	REDUCE (single alternator)
Operating engine	MONITOR – power as required, cowl flap as required, mixture adjusted
Fuel crossfeed	AS REQUIRED to use fuel from the dead-engine side (level flight only)
Trim	ADJUST for straight flight
ATC	ADVISE – nearest suitable airport

Air Start (Unfeather)

Restart attempt with altitude and cause identified

Fuel selector	ON
Auxiliary fuel pump	LO
Throttle	1/4 OPEN
Mixture	FULL RICH
Magnetos	ON (both)
Propeller	FORWARD of feather detent to low RPM (accumulator unfeathers if fitted)
Starter	ENGAGE if propeller does not windmill

▶ Engine restarted?

Yes:

Throttle	1000 RPM until oil pressure / temperature normal, then as required
Alternator	ON
Auxiliary fuel pump	OFF
Mixture / cowl flap	AS REQUIRED

Deferred: Engine restarted in flight – land as soon as practical, cause investigation required. ..... **NOTED**

No:

Propeller	FEATHER
Mixture	IDLE CUTOFF

Complete Engine Securing.

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel selector	**ON (crossfeed as required)**
Operating engine auxiliary fuel pump	**OFF (LO if required)**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Cowl flap	**AS REQUIRED**
Airspeed	**101 KIAS (Vyse) until landing assured**
Approach	**NORMAL glidepath or slightly high – avoid low, flat approaches**
Landing gear	**DOWN when landing assured (within gliding distance)**
Flaps	**APPROACH when landing assured; DOWN on short final only**
Airspeed	**REDUCE below 101 KIAS only on short final**

▶ Go-around required?

No:

Landing	**NORMAL – brakes as required**

Yes – above decision height:

Operating engine	**FULL POWER**
Flaps	**UP**
Landing gear	**UP (positive climb)**
Airspeed	**101 KIAS (Vyse)**
Bank	**2–5° toward operating engine**

A single-engine go-around below approx 50 ft AGL, or once full flaps are extended, should not be attempted.

Engine Fire in Flight

LAND IMMEDIATELY

Affected engine fuel selector (memory)	**OFF**
Affected engine mixture (memory)	**IDLE CUTOFF**
Affected engine propeller (memory)	**FEATHER**
Affected engine auxiliary fuel pump (memory)	**OFF**
Affected engine throttle	**CLOSED**
Affected engine magnetos	**OFF**
Affected engine alternator	**OFF**
Affected engine cowl flap	**CLOSED**
Cabin heater	**OFF**
Airspeed	**101 KIAS or higher to help extinguish fire**
Landing	**NEAREST SUITABLE AIRPORT – do not attempt restart**

Engine Fire During Start

Starter (memory)	**CONTINUE CRANKING**

▶ Engine starts?

Yes:

Throttle	**1700 RPM for a few minutes**
Engine	**SHUT DOWN and inspect for damage**

No:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Auxiliary fuel pump	**OFF**
Fuel selector	**OFF**
Starter	**CONTINUE CRANKING to draw fire into engine**
Magnetos / battery / alternators	**OFF**
Fire extinguisher	**OBTAIN / USE**
Aircraft	**EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

- Battery & alternator switches (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heater (memory) ..... **OFF**
- Cabin vents ..... **CLOSED**
- Fire extinguisher ..... **ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... **CHECK – do not reset faulted**
- Battery switch ..... **ON**
- Alternators / avionics / equipment ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Landing gear and flaps are electric: plan a hand-crank gear extension and a flaps-up landing.

- Deferred: Electrical power off – manual gear extension, flaps-up landing, no landing lights. ..... **NOTED**

Emergency Descent

- Throttles (memory) ..... **CLOSED**
- Propellers (memory) ..... **HIGH RPM**
- Landing gear (memory) ..... **DOWN (below 152 KIAS)**
- Flaps (memory) ..... **APPROACH (below 152 KIAS)**
- Airspeed (memory) ..... **152 KIAS**
- Mixtures ..... **ENRICH as required**

Manual Gear Extension (Hand Crank)

Gear does not extend electrically

- Airspeed ..... **100 KIAS or less**
- Landing gear motor circuit breaker ..... **PULL**
- Landing gear switch ..... **DOWN**
- Hand crank ..... **REMOVE from stowage and ENGAGE (floor, behind / between front seats)**
- Hand crank ..... **TURN COUNTERCLOCKWISE to the stop (approx 50 turns)**
- Landing gear indicator ..... **GREEN – gear down**
- Hand crank ..... **DISENGAGE & STOW**

Do not reset the gear motor circuit breaker or move the gear switch in flight after a manual extension. Leave the gear down.

▶ Green light obtained?

Yes:

- Deferred: Gear extended manually – do not retract; gear system inspection required. ..... **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

- Runway ..... **LONGEST HARD SURFACE available, emergency services advised**
- Passenger briefing ..... **COMPLETE (belts tight, brace, exits)**
- Landing gear ..... **UP (or as it is – do not cycle)**
- Flaps ..... **DOWN**
- Approach ..... **NORMAL, 90–95 KIAS**
- Auxiliary fuel pumps ..... **OFF**
- Cabin door ..... **UNLATCHED**

On short final:

- Mixtures ..... **IDLE CUTOFF**
- Fuel selectors ..... **OFF**
- Magnetos ..... **OFF**
- Battery & alternator switches ..... **OFF**
- Touchdown ..... **MINIMUM SPEED, slightly nose high, wings level**
- Aircraft ..... **EVACUATE once stopped**